



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
07-03-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

PAPER-2

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

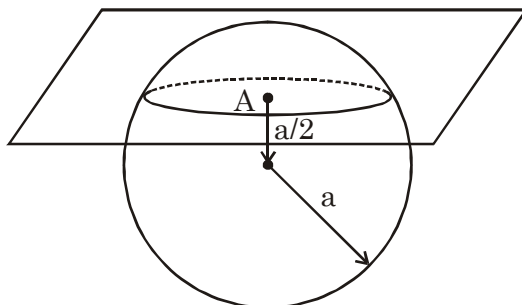
SECTION-I(i) : (Maximum Marks : 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. A disc of radius 1m is placed on the horizontal ground. It is uniformly charged with charge density of $3.54 \times 10^{-11} \text{ C/m}^2$. At a distance of $\frac{4}{3} \text{ m}$ from the centre we place a charge ($Q = 10\mu\text{C}$, $m = 10^{-6} \text{ kg}$) vertically above the centre. When we release it,
 - (A) It's initial acceleration is 3 m/s^2 upwards.
 - (B) It's initial acceleration is 6 m/s^2 downwards.
 - (C) It's equilibrium is at a height of $\frac{5}{\sqrt{3}} \text{ m}$ from the center.
 - (D) It's equilibrium is at a height of $\frac{1}{\sqrt{3}} \text{ m}$ from the center.

Space for Rough Work

2. An infinite plane cuts a spherical shell of radius a at a distance of $\frac{a}{2}$ from the centre. One of them is uniformly charged.



- (A) If the spherical shell was uniformly charged with charge density σ , the flux through the plane is $\frac{\sigma\pi a^2}{\epsilon_0}$
- (B) If the plane was uniformly charged with charge density σ , the flux through the shell is $\frac{3\pi\sigma a^2}{4\epsilon_0}$.
- (C) If the shell was uniformly charged, the potential at A (center of plane) will be maximum and decreases as we go out from point A.
- (D) If the plane was uniformly charged, the potential at top of shell is same as that at the center of the shell.
3. Inside a thin shell hollow metal sphere of radius 20 cm there is a metal ball of radius 10 cm. The centres of the ball and the sphere coincide. The inside ball is earthed with a very long wire, which is led through a small hole in the outer sphere. The outer sphere is given a charge of 10^{-8} C.
- (A) The potential of the outer sphere is 225 V
- (B) The potential of the inner sphere is 0
- (C) The charge on the inner surface of the outer sphere is 0.5×10^{-8} C
- (D) The charge on the outer surface of the outer sphere is 10^{-8} C

Space for Rough Work

4. Suppose the coloumb's law is defined as $\vec{F}_{12} = \frac{Kq_1q_2}{r_{12}^3} \hat{r}_{21}$ (Force on 2 due to 1). Then which of the following statement(s) will be **CORRECT** ?
- (A) Gauss theorem is still valid
 (B) The electric potential due to a positive charge increases with distance from that positive charge.
 (C) Like charges would repel
 (D) Like charges would attract
5. A satellite is put into a circular orbit at radial distance R_0 from the center of the Earth. The thin upper atmosphere of the Earth produces a viscous drag such that the satellite's orbit is observed to decay at a very slow rate. By slow we mean that the loss of energy per orbital period is small compared to the satellite's total kinetic energy. Moreover, it is observed that the rate of change of the radial distance dR/dt is constant. $dR/dt = -C$ where $C > 0$. If the viscous drag force on the satellite has the form $F(v) = Av^\alpha$, obtain expressions or constraints on the possible values of A and α .
- (A) $\alpha = -1$ (B) $A = \frac{GMmC^2}{2}$ (C) $A = \frac{mC}{2GM}$ (D) $\alpha = 3$
6. Two blocks of mass m_1 and m_2 ($m_1 > m_2$) are suspended by springs. When the blocks are attached to springs it is observed that equilibrium extension of each spring is same. Both the blocks have same amplitude of oscillation. Mark the correct option.
- (A) Both the blocks have same time period of oscillation.
 (B) Energy of oscillation of both the blocks is same.
 (C) Energy of oscillation of m_1 is greater.
 (D) information insufficient to answer.

Space for Rough Work

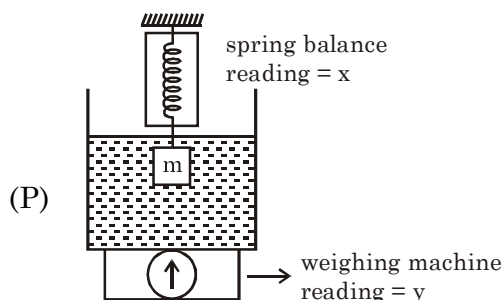
SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

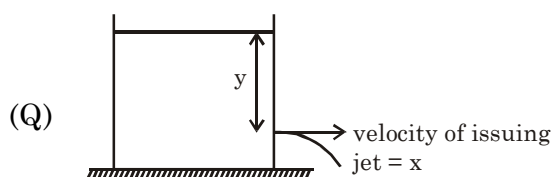
7. List-I shows certain situations and list-II contains effect on quantities x & y when the set up begins to fall freely. Match the list.

List-I

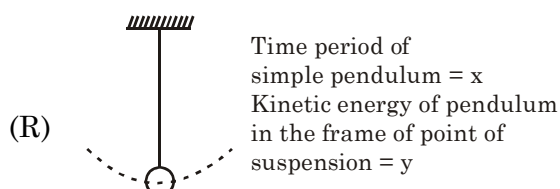
List-II



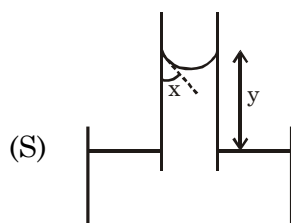
(1) x remains same



(2) y remains same



(3) x decreases



(4) y decreases

(5) x increases

(6) y increases

(A) $P \rightarrow 3,4$; $Q \rightarrow 1,2$; $R \rightarrow 1,2$; $S \rightarrow 3,6$

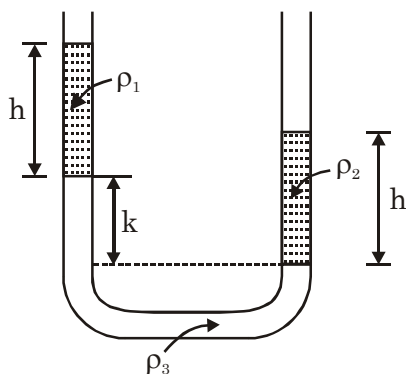
(B) $P \rightarrow 3,4$; $Q \rightarrow 2,3$; $R \rightarrow 2,5$; $S \rightarrow 1,6$

(C) $P \rightarrow 2,3$; $Q \rightarrow 1,4$; $R \rightarrow 2,5$; $S \rightarrow 4,5$

(D) $P \rightarrow 3,4$; $Q \rightarrow 2,3$; $R \rightarrow 2,5$; $S \rightarrow 5,6$

Space for Rough Work

8. Three liquids of densities ρ_1 , ρ_2 and ρ_3 and heights as shown are in equilibrium in a U-tube. Match the lists.



List-I

(P) $\rho_3 = 2(\rho_2 - \rho_1)$

(Q) $\rho_3 = \frac{(\rho_2 - \rho_1)}{2}$

(R) $\rho_1 = \rho_2$

(S) $\rho_2 = \frac{3\rho_1 + \rho_3}{3}$

(A) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

(C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$

List-II

(1) $k = 2h$

(2) $k = h/2$

(3) $k = 0$

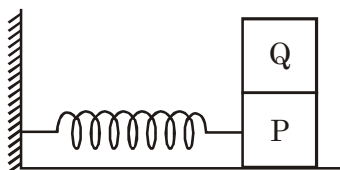
(4) None of the above

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(D) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 1$

Space for Rough Work

9. A block P of mass m is placed on smooth horizontal surface and connected with a spring. Another block Q of same mass is placed on the block P. The two blocks are pulled by a distance A and released. The blocks oscillates without slipping on each other. Then match the matrix.



List-I

- (P) Friction force on the block Q, when both the blocks coming towards equilibrium position from left.
- (Q) Friction force on the block Q when the block going right from equilibrium position. (exclude the equilibrium position)
- (R) Friction force will be maximum on Q
- (S) Friction force on P when both blocks are coming towards the equilibrium position from right.
- (A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$
- (C) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 5$

List-II

- (1) towards right
- (2) towards left
- (3) at equilibrium position
- (4) at extreme position
- (5) none of these
- (B) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 5$
- (D) $P \rightarrow 5; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

Space for Rough Work

10. Match the appropriate energy stated of list-I with their values in list-II.

List-I		List-II
(P) Self energy of a conducting shell with total charge Q	(1)	$\frac{KQ^2}{2R}$
(Q) Self energy of a uniform volumetric charged sphere having charge Q	(2)	$\frac{5KQ^2}{R}$
(R) Potential energy of three point charges Q, 2Q & Q placed at vertices of equilateral triangle of side R	(3)	$\frac{3KQ^2}{5R}$
(S) Potential energy of two point charges Q & $\frac{Q}{2}$ placed at a separation R	(4)	0
(A) P → 3; Q → 1; R → 2; S → 1	(B) P → 1; Q → 3; R → 2; S → 1	
(C) P → 1; Q → 1; R → 3; S → 4	(D) P → 3; Q → 3; R → 4; S → 1	

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

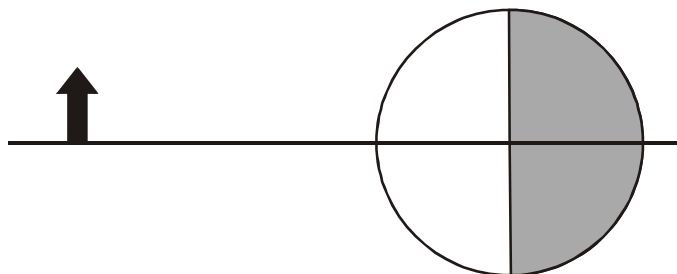
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

										+	
				0	0	0	0	0	0		
1	1	1	1	1	1	1	1	1	1		
2	2	2	2	2	2	2	2	2	2		
3	3	3	3	3	3	3	3	3	3		
4	4	4	4	4	4	4	4	4	4		
5	5	5	5	5	5	5	5	5	5		
6	6	6	6	6	6	6	6	6	6		
7	7	7	7	7	7	7	7	7	7		
8	8	8	8	8	8	8	8	8	8		
9	9	9	9	9	9	9	9	9	9		

										-	
				0	0	0	0	0	0		
1	1	1	1	1	1	1	1	1	1		
2	2	2	2	2	2	2	2	2	2		
3	3	3	3	3	3	3	3	3	3		
4	4	4	4	4	4	4	4	4	4		
5	5	5	5	5	5	5	5	5	5		
6	6	6	6	6	6	6	6	6	6		
7	7	7	7	7	7	7	7	7	7		
8	8	8	8	8	8	8	8	8	8		
9	9	9	9	9	9	9	9	9	9		

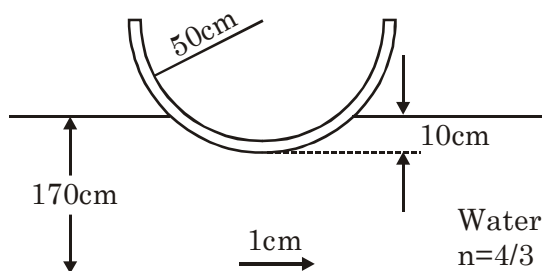
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A glass ball of radius 10 cm has an index of refraction 1.5. The back half of the ball is silvered so that it acts as a concave mirror as shown in figure. Find the position of the final image (in cm) seen by an eye to the left of the object and ball for an object at a distance of 30 cm to the left of the front surface of the ball. Write distance from left face of the ball.



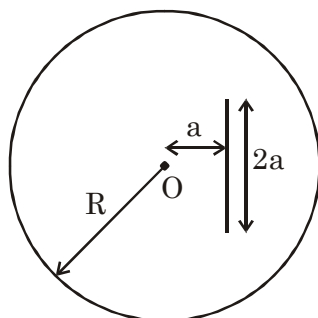
Space for Rough Work

2. Find the magnitude of electric field (in N/C) due to a dipole of dipole moment 1 nCm placed at origin. The electric field at $(1, 2, 0)$ is to be found.
3. A fish is rising up vertically inside a pond with velocity of 4 cm/s , and notices a bird, which is diving vertically downward and its velocity appears to be 16 cm/s (to the fish). What is the real velocity (in cm/s) of the diving bird, if refractive index of water is $4/3$.
4. A thin hemispherical bowl of transparent plastic floats on water in a tank. The radius of the bowl is 50 cm and the depth of the bowl in water is 10 cm . The depth of the water ($\mu = 1.33$) in the tank is 170 cm . An object 1 cm long is on the bottom of the tank directly below the bowl. The object is viewed from directly above the bowl. Ignore the refractive effects of the plastic. What is the size (in cm) of the image?

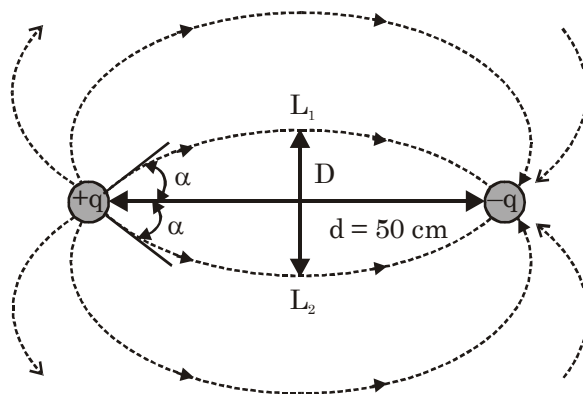


Space for Rough Work

5. Consider a horizontal charged disk with center O and having surface charge density σ , and radius R, Consider a vertical rectangle of dimensions $2a \times 2h$, which cuts the disk at a distance 'a' from its center, such that half of rectangle lies below the disk and half of it lies above the disk. Use $a \ll R$ to find the value of flux of electric field (in $\mu\text{Nm}^2/\text{C}$) through the rectangle to the nearest integer. The top view is shown below:
(Take : $h = 3\text{cm}$, $R = 4\text{ cm}$, $a = 4\text{ }\mu\text{m}$, $\sigma = 8.85\text{ }\mu\text{C}/\text{m}^2$)

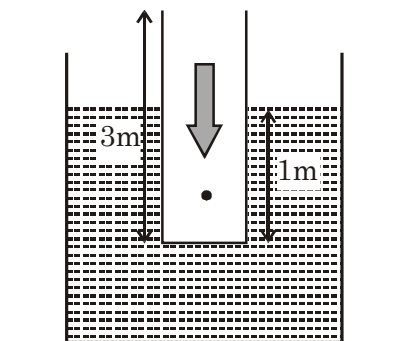


6. Two charges $+q$ and $-q$ are separated by a distance d . The electric field lines for this system form the well-known pattern showed in the figure below. Consider two lines and lying on the same plane and leaving the positive charge at $\alpha = 60^\circ$ each. What is the maximum distance between the lines in centimeters.



Space for Rough Work

7. A cylindrical pipe of outer cross sectional area A is submerged in liquid as shown. The mass of the pipe is M . A particle of the same mass collides inelastically with the floor of the pipe. The pipe just gets completely submerged in the liquid after the collision takes place. What is the velocity (in m/s) of the particle before the collision ? Make all reasonable assumptions. The size of the container is very large.



8. A pendulum bob is hung from a string of length 1m and is swinging in a vertical plane with an angular amplitude of 1.8° . what is its acceleration when it passes through the bottom most point (in cm/s^2)? (Take : $\pi^2 = 10$)

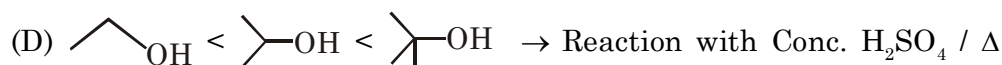
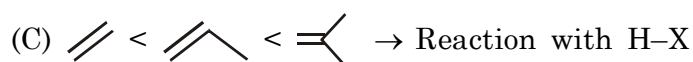
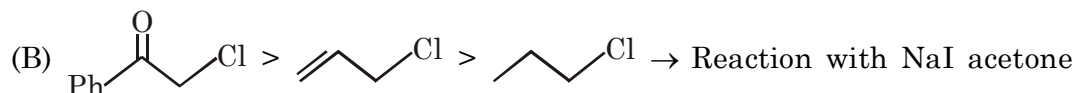
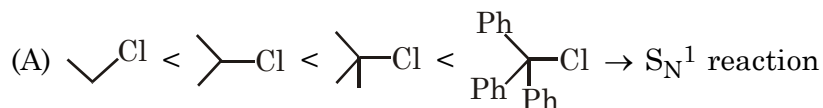
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks : 24)

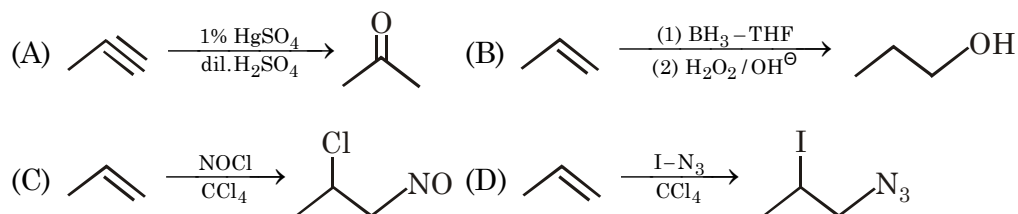
- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Identify correct reactivity order ?

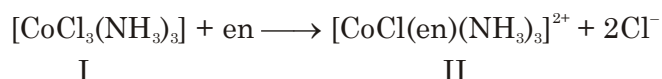


Space for Rough Work

2. Identify reactions correctly matched with their products ?



3. The **CORRECT** statement(s) for the following reaction is/are.

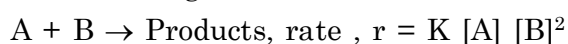


- (A) If complex compound-I is a fac isomer, then complex ion-II will be fac-isomer w.r.t. NH_3 .
 (B) If the complex compound-I is mer isomer, then two isomers are produced.
 (C) If complex compound-I is a fac isomer, then complex ion-II exhibit optical activity.
 (D) If complex compound-I is a mer isomer then complex ion-II does not exhibit optical activity
4. Find out **CORRECT** IUPAC name(s) of complex compounds.
- (A) Pentaamminecyanidoiron(II) hexanitrito-N-irridate(III)
 (B) Triamminetricyanidochromium(III) hexanitrito-N-irridate(III)
 (C) Hexanitrito-N-irridium(III) pentaamminecyanidochromate(II)
 (D) Pentaamminecyanidoiron(III) hexanitrito-N-irridate(III)

Space for Rough Work

5. The reaction : $A(aq.) \rightarrow 2B(aq.) + C(aq.)$ follows first order kinetics. The osmotic pressure of solution was 2.0 atm at $t = 0$ and 3.0 atm at $t = 15$ min, at constant temperature (300K). The correct information(s) about the reaction is /are -
($\ln 2 = 0.7$, $\ln 3 = 1.1$, $R = 0.08$ L-atm/K-mol.)
- (A) 25% reaction is completed in 15 min.
(B) The rate constant of reaction is 0.02 min^{-1}
(C) After a long time, the limiting osmotic pressure of solution becomes 4.5 atm.
(D) The rate of reaction at $t = 15$ min. is $1.25 \times 10^{-3} \text{ M min}^{-1}$

6. The following data is obtained at 25°C for the reaction :



$[A_0] / \text{M}$	$[B_0] / \text{M}$	$[t_{1/2}] / \text{min.}$
0.8	0.002	10
0.4	0.002	x
0.2	0.004	y
0.004	0.9	20
0.008	0.45	z

The correct value(s) of x, y and z is/are -

- (A) $x = 20$ (B) $y = 20$ (C) $z = 20$ (D) $x = 10$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

7.	List-I (Reactions)	List-II (Parameters, effectively favouring the reaction in forward direction)
	(P) $2\text{NO(g)} \rightleftharpoons \text{N}_2\text{(g)} + \text{O}_2\text{(g)} ; \Delta H^0 = -\text{ve}$	(1) Increase in temperature
	(Q) $\text{PCl}_5\text{(g)} \rightleftharpoons \text{PCl}_3\text{(g)} + \text{Cl}_2\text{(g)} ; \Delta H^0 = +\text{ve}$	(2) Increase in pressure
	(R) $2\text{A(aq.)} \rightleftharpoons 3\text{B(aq.)} + \text{H}_2\text{O(l)} ; \Delta H^0 = +\text{ve}$	(3) Increase in volume of system
	(S) $\text{CaCO}_3\text{(s)} \rightleftharpoons \text{Ca}^{2+}\text{(aq.)} + \text{CO}_3^{2-}\text{(aq.)} ; \Delta H^0 = +\text{ve}$	(4) Increase in amount of water (only for reaction in aq. form)
		(5) Increase in moles of reactants at constant volume

The correct option is -

- (A) P → 5 ; Q → 1, 3, 5; R → 1, 3, 4, 5 ; S → 1,3,4
 (B) P → 1,2,5 ; Q → 1, 2; R → 1, 4, 5 ; S → 1, 4
 (C) P → 5 ; Q → 1, 3, 5; R → 1, 3, 5 ; S → 1,2,5
 (D) P → 5 ; Q → 3, 5; R → 1, 2, 3 ; S → 1,3,4

Space for Rough Work

8. List-I

- (P) $\text{Na}_{1.00} \text{Cl}_{0.97}$
 (Q) $\text{Zn}_{1.08} \text{O}_{1.00}$
 (R) $\text{Ni}_{0.98} \text{O}_{1.00}$
 (S) $\text{Fe}_{0.95} \text{O}_{1.00}$

List-II

- (1) Yellow colour
 (2) $d_{\text{experimental}} < d_{\text{ideal crystal}}$
 (3) $d_{\text{experimental}} > d_{\text{ideal crystal}}$
 (4) p-type semiconductor
 (5) Metal excess defect due to excess cation
 (6) Metal excess defect due to anion vacancy

The correct option is -

- (A) $P \rightarrow 1,3$; $Q \rightarrow 1, 2, 5$; $R \rightarrow 1, 4, 5$; $S \rightarrow 1,2,5$
 (B) $P \rightarrow 1,2,6$; $Q \rightarrow 1, 5$; $R \rightarrow 1, 2,4$; $S \rightarrow 2, 4$
 (C) $P \rightarrow 1,2,6$; $Q \rightarrow 1, 3, 5$; $R \rightarrow 2, 4$; $S \rightarrow 2,4$
 (D) $P \rightarrow 1,2,4$; $Q \rightarrow 1, 3, 5$; $R \rightarrow 1, 2, 4$; $S \rightarrow 2, 4$

9. Match the list : (Atomic number of Ni = 28, Ir = 77, Pt = 78, Co = 27)

List- I

(Co-ordination compounds)

- (P) $\text{K}_2[\text{NiBr}_2(\text{NO}_3)_2(\text{en})]$
 (Q) $[\text{IrCl}(\text{SCN})(\text{H}_2\text{O})(\text{NH}_3)_3]$
 (R) $[\text{Pt}(\text{gly})_2(\text{NO}_2)_2]$
 (S) $[\text{Co}(\text{gly})(\text{H}_2\text{O})_2(\text{NH}_3)_2][\text{Cu}(\text{NO}_2)_4]$

List- II

(Characteristics of coordination compounds)

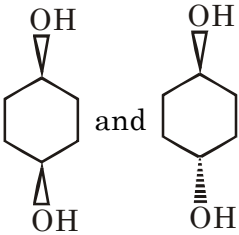
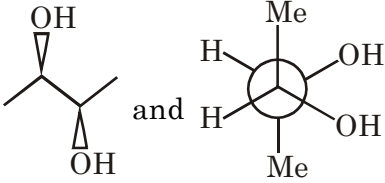
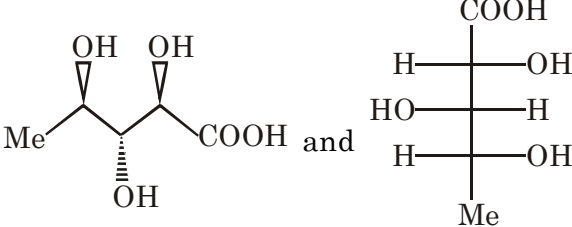
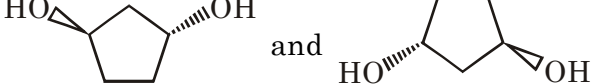
- (1) It exhibit structural isomerism
 (2) High spin complex
 (3) Low spin complex
 (4) Electron migrates from (n-1)d subshell of central metal cation to its higher energy subshell
 (5) Total number of optically active isomers are equal to two.

The correct option is

- (A) $P \rightarrow 5$; $Q \rightarrow 1,3,4,5$; $R \rightarrow 1,3$; $S \rightarrow 1,3,4$
 (B) $P \rightarrow 2,5$; $Q \rightarrow 3,4,5$; $R \rightarrow 3,5$; $S \rightarrow 1,2,5$
 (C) $P \rightarrow 2,4$; $Q \rightarrow 3,4,5$; $R \rightarrow 1,3,4$; $S \rightarrow 1,4$
 (D) $P \rightarrow 1,2$; $Q \rightarrow 3,5$; $R \rightarrow 1,2$; $S \rightarrow 1,4,5$

Space for Rough Work

10. Match the list

List-I Pair of compounds	List-II Relation
(P) 	(1) Diastereo isomers
(Q) 	(2) Enantiomers
(R) 	(3) Geometrical isomers
(S) 	(4) Identical

The correct option is :

- (A) P $\rightarrow$ 1,3; Q $\rightarrow$ 2; R $\rightarrow$ 1; S $\rightarrow$ 4
 (B) P $\rightarrow$ 3; Q $\rightarrow$ 1, 2; R $\rightarrow$ 2; S $\rightarrow$ 4
 (C) P $\rightarrow$ 1,2; Q $\rightarrow$ 3; R $\rightarrow$ 4; S $\rightarrow$ 1
 (D) P $\rightarrow$ 1,3,4; Q $\rightarrow$ 2,4; R $\rightarrow$ 1; S $\rightarrow$ 4

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

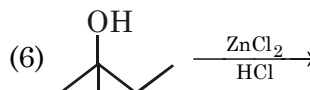
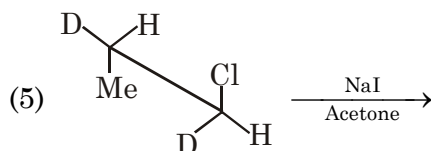
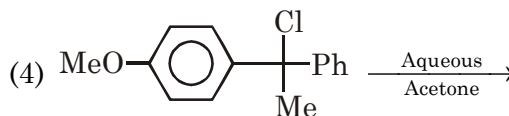
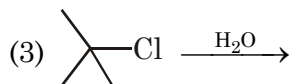
- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+					-				
●	●	●	●	●	●	●	●	●	
①	①	①	①	①	①	①	①	①	
②	②	②	②	②	②	②	②	②	
③	③	③	③	③	③	③	③	③	
④	④	④	④	④	④	④	④	④	
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	

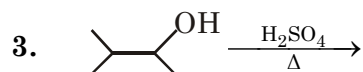
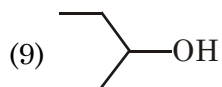
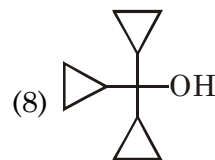
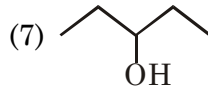
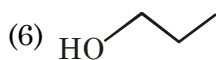
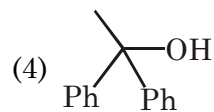
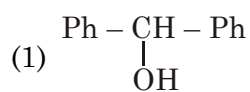
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Identify total number of S_N^2 reactions?



Space for Rough Work

2. Identify total number of alcohols gives immediate white turbidity with $(\text{ZnCl}_2 + \text{HCl})$ Lucas reagent?



Identify total number of ' α ' H in major product of above reaction?

Space for Rough Work

-
4. Find out total number of molecules which can undergo hydrolysis at room temperature and hybridization of central atom is sp^3d in transition state.
 CCl_4 , $SiCl_4$, PCl_3 , $AsCl_3$, SF_6 , SeF_6
5. Find out maximum number of electrons of Ni^{2+} ion which are present in non axial d-orbitals in the complex compound : $[Ni(dmg)_2]$
6. If EAN of central metal ion X^{2+} in a non chelating complex is 34 and atomic number of X is 28. The number of monodentate ligands in complex are :
-

Space for Rough Work

7. 25 gm $\text{CaCO}_3(\text{s})$ is taken in an empty vessel of 16.42 L capacity and heated to 627°C . The mass percent of $\text{CaCO}_3(\text{s})$ decomposed is -
 $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$; $K_p = 0.9 \text{ atm}$ at 627°C .
(Given : $R = 0.0821 \text{ L-atm/K-mol}$)
8. 12.5 ml O_2 gas dissolves in 200 gm water at 25°C and 4 atm. The volume (in ml) of O_2 gas dissolved in 100 gm water at 25°C and 2 atm is.

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks : 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following function(s) is/are differentiable at $x = 0$?
 (A) $f(x) = |x^3|$ (B) $f(x) = x^3|x|$ (C) $|x| \sin^3 x$ (D) $f(x) = x|\tan^3 x|$
2. If $f(x) = \begin{cases} 2x^2 - 3x + 2a & ; x \in \mathbb{Q} \\ x^2 - bx & ; x \notin \mathbb{Q} \end{cases}$ is continuous at $x = 1$ and $x = 3$, then
 (A) $a = -1$ (B) $a = \frac{3}{2}$ (C) $b = -1$ (D) $b = \frac{3}{2}$
3. If α, β are roots of $9x^2 - 11x + 1 = 0$ then value of $\frac{1}{(9\alpha - 11)^2} + \left(\frac{11\beta - 1}{9}\right)$ is-
 (A) $\frac{56}{47}$ (B) $\frac{67}{56}$ (C) $\frac{81}{67}$ (D) $\frac{103}{81}$

Space for Rough Work

4. If $\frac{\cos^4 x}{\theta_1} + \frac{\sin^4 x}{\theta_2} = \frac{1}{(\theta_1 + \theta_2)}$, then $\frac{d\theta_2}{d\theta_1}$ equals (where x is not dependent on θ_1 or θ_2)
- (A) $\sin^2 x$ (B) $\cos^2 x$ (C) $\tan^2 x$ (D) $\sec^2 x$
5. A right angled triangle has one leg of length 1, another of length ' x ' & hypotenuse of length ' y '. The angle opposite to the side x is θ then -
- (A) $y^n - x^n = \frac{1 - \sin^n \theta}{\cos^n \theta}$ (B) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y - x) = 0$
- (C) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y^2 - x^2) = 1$ (D) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y^3 - x^3)$ does not exist.
6. Value of $\lim_{x \rightarrow 1} \frac{(\log(1+x) - \log 2)(3 \cdot 4^{x-1} - 3x)}{((7+x)^{1/3} - (1+3x)^{1/2}) \sin \pi x}$ is equal to
- (A) $\frac{9}{\pi}(2\log 2 - 1)$ (B) $\frac{9}{4\pi}[\log 4 - 1]$ (C) $\frac{9}{2\pi}\left[\log 4 - \frac{1}{2}\right]$ (D) $\frac{2}{3\pi}[2\log 2 - 1]$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 12)

- This section contains **FOUR** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

7. Let $f(x) = x^3 - 6x^2 - 36x + 2$

List-I

List-II

- (P) $f(x)$ is increasing function in (1) $\left(0, \frac{3}{2}\right)$
- (Q) $x = C$ is point of inflection for curve $y = f(x)$, then (2) $\left(\frac{3}{2}, 4\right)$
 C lies in
- (R) Let N denotes number of real root(s) of equation (3) (5,7)
 $f(x) = 0$, then N lies in
- (S) Let Rolle's theorem holds for $f(x)$ in $[0, a]$. If k is (4) (8,10)
sum of all possible values of a , then k lies in (5) (9,11)
- (A) $P \rightarrow 4, 5$; $Q \rightarrow 2$; $R \rightarrow 1$; $S \rightarrow 4$ (B) $P \rightarrow 4, 5$; $Q \rightarrow 2$; $R \rightarrow 2$; $S \rightarrow 4, 5$
- (C) $P \rightarrow 5$; $Q \rightarrow 1$; $R \rightarrow 2$; $S \rightarrow 4, 5$ (D) $P \rightarrow 4$; $Q \rightarrow 3$; $R \rightarrow 2$; $S \rightarrow 5$

8. **List-I**

List-II

- (P) If $y = |\ln(x^2)|$ then $y'\left(-\frac{1}{2}\right)$ is equal to (1) 4
- (Q) If $f(x) = \frac{1}{2}x^3 + \tan^{-1}x - \frac{\pi}{4}$ and $g(x) = f^{-1}(x)$, (2) 5
then $\left|16g''\left(\frac{1}{2}\right)\right|$ is equal to (3) 2
- (R) If $f(x) = x^{x^x}$, then $f''(1)$ is equal to (4) 1
- (S) If $\ln((e-1)e^{xy} + x^2) = x^2 + y^2$, then $\frac{dy}{dx}\bigg|_{(1,0)}$ is (5) Harmonic mean of 3 and $\frac{3}{2}$
- (A) $P \rightarrow 4$; $Q \rightarrow 2$; $R \rightarrow 5$; $S \rightarrow 3, 5$ (B) $P \rightarrow 1, 5$; $Q \rightarrow 2$; $R \rightarrow 3$; $S \rightarrow 5$
- (C) $P \rightarrow 1$; $Q \rightarrow 2$; $R \rightarrow 3, 5$; $S \rightarrow 3, 5$ (D) $P \rightarrow 4$; $Q \rightarrow 1$; $R \rightarrow 3, 5$; $S \rightarrow 3, 5$

Space for Rough Work

9. List-I

List-II

- (P) Value of λ for which the $f(x) = 2x^3 - 3(\lambda + 2)x^2 + 12\lambda x$ has one local maxima and one local minima in $\mathbb{R}$, can be (1) 0
- (Q) If $f(x) = x^{\frac{1}{\ln x}}$ then $f'(x) = 0$ for x equals to (2) 1
- (R) Cosine of the angle of intersection of curves $f(x) = 2^x \ln x$ and $g(x) = x^{2x} - 1$ is (3) 2
- (S) Consider curve $y^3 - 3xy + 2 = 0$ then total no of horizontal & vertical tangent is (4) 3
- (5) e
- (A) $P \rightarrow 1, 2, 4, 5$; $Q \rightarrow 3, 4, 5$; $R \rightarrow 2$; $S \rightarrow 3$ (B) $P \rightarrow 2, 4, 5$; $Q \rightarrow 2, 3, 4, 5$; $R \rightarrow 2$; $S \rightarrow 3$
- (C) $P \rightarrow 1, 2, 4, 5$; $Q \rightarrow 3, 4, 5$; $R \rightarrow 3$; $S \rightarrow 2$ (D) $P \rightarrow 1, 2, 4, 5$; $Q \rightarrow 2, 3, 4, 5$; $R \rightarrow 3$; $S \rightarrow 2$

10. Let $f(x) = \begin{cases} x+a, & x < 0 \\ \sqrt{x^2 - 2x + 1}, & x \geq 0 \end{cases}$

and $g(x) = \begin{cases} x+1, & x < 0 \\ x^2 + 2(b-x) + 1, & x \geq 0 \end{cases}$

a, b are non-negative real numbers. If $(g \circ f)(x)$ is continuous for all real x

List-I

List-II

- (P) Number of possible values of b is (1) 0
- (Q) Sum of all possible values of a is (2) 1
- (R) Value of $\frac{d(g(f(x)))}{dx}$ at $x = 4$ is (3) 2
- (S) If $f(x)$ is non differentiable at k points, then the value of k is (4) 4
- (5) greater than 4
- (A) $P \rightarrow 1$; $Q \rightarrow 2$; $R \rightarrow 4$; $S \rightarrow 3$ (B) $P \rightarrow 1$; $Q \rightarrow 2$; $R \rightarrow 5$; $S \rightarrow 3$
- (C) $P \rightarrow 3$; $Q \rightarrow 1$; $R \rightarrow 5$; $S \rightarrow 4$ (D) $P \rightarrow 3$; $Q \rightarrow 2$; $R \rightarrow 4$; $S \rightarrow 3$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+											
●	●	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

-											
●	●	●	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- The ordinate of a point describing the circle $x^2 + y^2 = 25$ decreases at the rate of 1 cm/sec, then the rate of change (in cm/sec) of abscissa of the point when ordinate equal to 3cm, is-
(Given $x > 0$, $y > 0$)
- Let $f(x) = x + 1$, $g^{-1}(x) = x^3 + x + 1$ and $h(x) = g(f(x))$, then $\lim_{x \rightarrow 1} \frac{h^{-1}(x) - h^{-1}(1)}{x - 1}$ is equal to
- The maximum value of $f(x) = \frac{2x^2 + 7x + 3}{2x^2 + x + 1}$, is

Space for Rough Work

4. Let $P(x) = x^{10} + ax^8 + bx^6 + cx^4 + dx^2 - 1$ be polynomial with real coefficients. If $P(1) = 1$, $P(2) = -1$ then the minimum number of real zeroes of $P(x)$ is
5. Consider the equation $x^2 + 2x - n = 0$, where $n \in \mathbb{N}$ and $n \in [5, 100]$. Total number of different values of 'n' so that the given equation has integral roots, is
6. Value of $\left| \lim_{x \rightarrow \frac{3\pi}{4}} \frac{4 \sin^2 x \cos x - \cos x + \sin x}{\sin x + \cos x} \right|$ is

Space for Rough Work

7. If $f(x) = \begin{cases} \frac{\sin^{-1}(\{x\}^2 + \{x\}^3) - x}{\{x\}}, & x \neq 0 \\ -1, & x = 0 \end{cases}$, then the absolute value of RHD of $f(x)$ at $x = 0$, is

(where $\{.\}$ denotes fractional part function)

8. Let $f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5}$ and $g(x) = f^{-1}(x)$, then $|g''(0)|$ is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
I(ii)	Matching Lists Type (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	12
II	Numerical Value Type (Up to second decimal place)	8	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator